

MAYA

Quantitative Research Infrastructure

Architecture, Research Methodology & HANNA Live Lifecycle

Methodology Document - Version 2.0

MAYA is conceived as a controlled research laboratory that moves market observations through discovery, strict validation, true out-of-sample evaluation and, after approval, live management through HANNA. HANNA is the operational layer that manages approved individuals without allowing live adaptation to bypass the research process.

1. Executive Philosophy

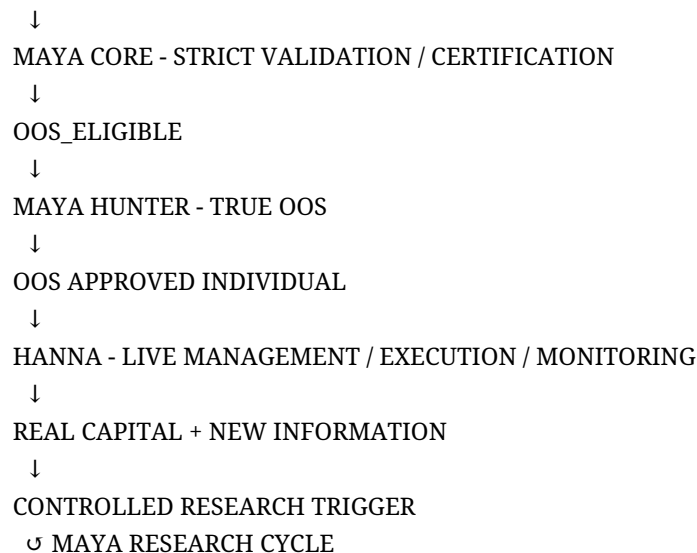
MAYA is not designed as a single trading robot and not as a machine that guarantees profitability. It is a research infrastructure whose purpose is to progressively answer a chain of questions:

Does an apparent pattern exist?
↓
Is it statistically defensible?
↓
Does it survive temporal separation?
↓
Is it economically exploitable?
↓
Can it be converted into an executable rule?
↓
Does it survive true out-of-sample conditions?
↓
Can it operate live under controlled risk and adaptation?

The original document defined the first four MAYA stages as Collector, EDA-Lite, MAYA Core and MAYA Hunter. This version extends that lifecycle with HANNA, which begins only after a candidate has passed the research gates and becomes an approved operational individual.

2. The Complete MAYA Lifecycle

MARKET
↓
MAYA COLLECTOR
↓
EDA-LITE - DISCOVERY / CALIBRATION
↓
STRONG_CANDIDATE



Layer	Main question	Primary output	Role
Collector	What happened in the market?	Structured event dataset	Evidence collection
EDA-Lite	Where might an edge exist?	Candidate / Strong Candidate	Discovery and triage
Core	Can the candidate survive a stricter protocol?	Certified / OOS Eligible	Validation and certification
Hunter	What happens on unseen data?	OOS evidence	True OOS evaluation
HANNA	How should an approved individual operate and adapt safely?	Live individual state	Execution, risk, monitoring, controlled adaptation

3. MAYA Collector - The Evidence Factory

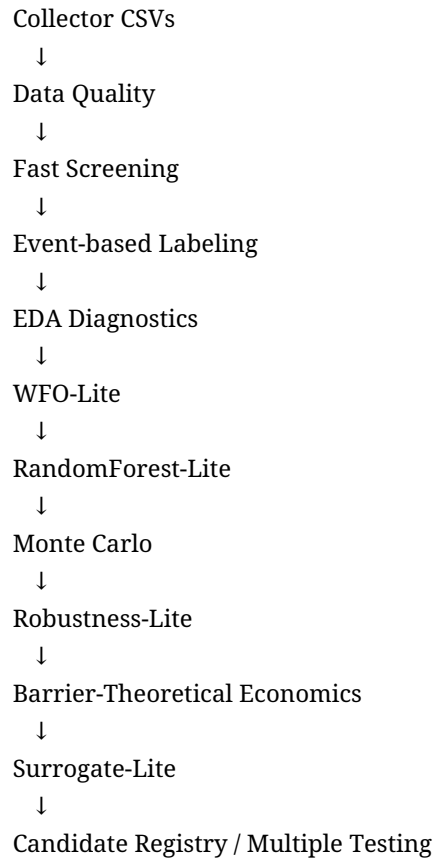
Collector is the market-data acquisition and feature-engineering layer. It does not certify strategies and does not decide whether an edge is real. Its function is to create a structured event-level research dataset from market activity.

- Capture event timing, fixing/range context and decision timing.
- Generate the predefined research features close to the data source.
- Record MFE/MAE barrier timing and data-quality status.
- Preserve the causal boundary: predictive inputs must be available at decision time.
- Provide a stable data contract to downstream research modules.

Mental model: Collector is the scientist collecting samples before the laboratory begins testing them.

4. EDA-Lite - Discovery and Calibration

EDA-Lite is the lightweight prospecting layer. It is intentionally less computationally complex than Core, but it keeps the temporal and causal controls that protect the research from obvious leakage.



The purpose is not to prove that a strategy will make money. Its purpose is to reduce the research universe and identify candidates that deserve the more expensive Core process.

5. MAYA Core - Strict Validation and Research Certification

Core receives a frozen candidate artifact rather than merely a performance score. It revalidates the candidate under a stricter protocol and determines whether the candidate is defensible under the declared MAYA certification rules.

- Reproduce the candidate definition from its artifact and data/code provenance.
- Apply stricter purged validation and model governance.
- Stress robustness across time, regimes and relevant parameter neighborhoods.
- Control multiple testing at the appropriate research family level.
- Evaluate economics under explicit and realistic cost assumptions.
- Audit reproducibility and rule/model integrity.
- Only then grant an OOS-eligibility state.

Certification does not mean guaranteed future profitability. It means that the candidate satisfied the requirements of the declared research protocol.

6. MAYA Hunter - True Out-of-Sample

Hunter receives an OOS-eligible candidate and must treat the target OOS period as evidence, not as a tuning set. The model/rule is frozen for the declared test period.

```
FROZEN CANDIDATE
↓
UNSEEN DATA
↓
EXECUTION ASSUMPTIONS
↓
TRADE-LEVEL RESULTS
↓
OOS SCORECARD
```

Hunter answers a simple question: does the candidate survive when it leaves the research laboratory?

7. The Transition from Hunter to HANNA

A key conceptual boundary is reached after Hunter: a candidate that passes the OOS gate becomes an approved operational individual. This is where HANNA begins.

```
EDA-Lite → STRONG_CANDIDATE
Core → CERTIFIED / OOS_ELIGIBLE
Hunter → OOS APPROVED INDIVIDUAL
HANNA → LIVE INDIVIDUAL
```

HANNA should not be thought of simply as another backtest or another model-selection engine. It is the live lifecycle manager for already-approved quantitative individuals.

8. HANNA - Philosophy of the Live Individual

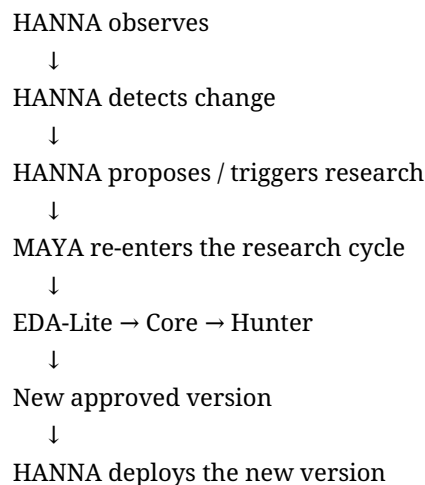
HANNA is the operational layer that manages a population of approved quantitative individuals. An individual combines an approved candidate, a model/rule version, an asset/profile identity, execution rules, risk constraints and a complete research provenance chain.

```
HANNA_001
Asset: XAUUSD
Profile: PRE_ASIA
Candidate: XU_PRE_ASIA_BARRIER15_0042
Model: MODEL_v001
Core certificate: CORE_CERT_0042
Hunter OOS: PASSED
Status: LIVE
```

In this example, HANNA(XU_PRE_ASIA) is the operational instance responsible for applying the frozen XAUUSD PRE-ASIA rule under its approved constraints.

9. HANNA Does Not Mean Uncontrolled Self-Modification

The most important governance principle for HANNA is: it may learn from new information, but it may not certify itself.



This preserves the separation between live evidence and research. A live loss should not directly cause an automatic rule change, because that would turn the live period into an uncontrolled tuning set.

10. Versioning of OOS Evidence

The same calendar period cannot remain pure OOS for every future version of the model. Once new live data are used to research a new model, those data become part of the research history for that new version.

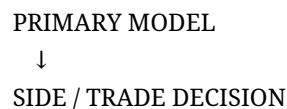
MODEL_v001
Train: 2016–2022
Internal Validation: 2023
True OOS: 2024–2026

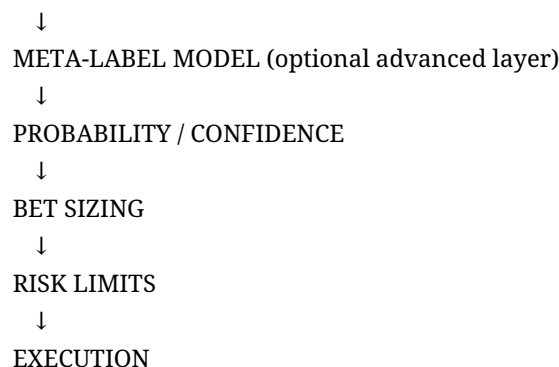
MODEL_v002
Can use 2024–2026 as research data
Therefore its future OOS must start after its new validation boundary

This is why model versioning and candidate provenance are central to HANNA.

11. Signal, Meta-Labeling and Bet Sizing

The primary model and capital allocation should be treated as separate layers. A primary model can answer whether a trade should be considered; a downstream risk layer can decide how much capital to allocate within approved constraints.





Meta-labeling is a natural future extension when the primary signal is already defensible and a second model is used to estimate whether the signal should be taken or how aggressively it should be sized. It should be researched and validated rather than introduced as an automatic live optimization mechanism.

Kelly or fractional Kelly can be considered as a position-sizing method. D'Alembert and similar progression schemes should be treated as money-management experiments, not as evidence that the underlying alpha exists.

12. What Does and Does Not Change the Model?

Layer / change	Does it modify the predictive model?	Governance
Position sizing / capital allocation	No, not by itself	Must stay within approved risk rules and be validated as a complete portfolio/risk layer
Temporary risk throttle	No	Operational adaptation allowed only within predefined limits
Threshold / feature / TP-SL change	Yes	New research candidate; return to MAYA
Model architecture change	Yes	New version; EDA-Lite/Core/Hunter cycle required
Meta-label model update	Potentially	Must be trained, validated and versioned

13. Performance Measurement - No Single Magic Number

MAYA/HANNA should not be reduced to maximizing a single OnTester number. Different metrics answer different questions.

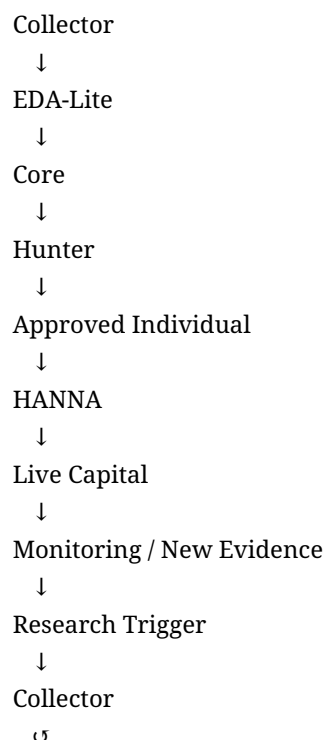
Dimension	Examples	Question answered
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Statistical edge	MCC, PR-AUC, calibration	Does the signal contain predictive information?
Economic edge	Expectancy, Profit Factor, cost-adjusted returns	Does the signal have economic value under declared assumptions?
Risk	Max drawdown, Sharpe, Sortino, CVaR	How is the return distributed relative to risk?
Robustness	Fold/year/regime stability, worst fold	Does the result depend on a narrow historical condition?
Selection control	Monte Carlo, multiple-testing records, DSR context	How much should we discount an apparently exceptional result?
Operations	Slippage, fills, latency, execution drift	Can the edge survive actual execution?

Deflated Sharpe Ratio can be used as a supporting selection-aware metric when appropriate, especially after a large research search. It is not a replacement for temporal validation, Monte Carlo, economic analysis or true OOS.

14. The Research Cycle Becomes a Controlled Feedback Loop

RESEARCH FACTORY



The loop is intentional, but it is not an uncontrolled self-learning loop. New live information can initiate a new research cycle, while historical OOS evidence remains immutable for the version it was used to evaluate.

15. The Four Levels of Edge

- Statistical edge: evidence of a predictive relationship under the declared statistical protocol.
- Economic edge: the relationship remains attractive after the declared economic assumptions and costs.
- Executable edge: the signal can be represented as a deterministic, auditable rule/model that Hunter can implement.
- Live edge: the approved individual continues to behave acceptably under live conditions.

A candidate can pass one level and fail another. A live failure does not imply that the research framework failed; it may show that the apparent edge was not persistent.

16. Final Philosophy of MAYA + HANNA

MAYA discovers, challenges, validates and tests.

HANNA deploys, manages, monitors and controls the live lifecycle.

Neither layer is allowed to bypass the research governance that protects the integrity of the system.

17. Architectural State to Carry Forward

01 MAYA COLLECTOR

Research data + feature engineering

02 MAYA EDA-LITE

Discovery / calibration / candidate triage

03 MAYA CORE

Strict validation / certification

04 MAYA HUNTER

Frozen true OOS evaluation

05 HANNA

Approved individual lifecycle: execution + risk + monitoring + controlled adaptation

The next engineering contract to define is the OOS Approved Individual Contract: the machine-readable artifact that Hunter hands to HANNA, containing the candidate identity, model/rule version, feature contract, Core certificate, OOS evidence, risk constraints, execution constraints, and the exact adaptations permitted in live operation.